Pair-HMM parameters from HiFi alignments

Stage G of `docs/phmm.md`: the four gap parameters of the pair HMM, estimated from reads aligned to the HG002 curated assembly. This report documents which alignments were used, what was counted in them, and what the counts imply.

Source: `hg002.curated.sorted.bam`, minimap2 2.30 `map-hifi`. Generated by `qamodel/estimate.py`; provenance in `docs/phmm_params_hg002.json`.

1. Choosing where to look

The assembly is tiled into non-overlapping 100 kb tiles and 40,000 of them are drawn **without replacement** from a seeded RNG, so a run reproduces from `--seed 0` alone. That is 4.0 Gb of the 6.0 Gb diploid assembly.

Sampling tiles rather than reading the head of the BAM is the point: gap rates vary several-fold with sequence context, and context is not uniformly distributed along an assembly. Reading the first N reads would give one region's opinion at full apparent precision.

The draws are without replacement because overlapping windows would count a read in the intersection twice, and would break the jackknife of section 5, which treats each tile as an independent block.

2. Choosing which alignments to count

Within a tile, every alignment touching it is offered, and rejected in this order. Secondary, supplementary and unmapped records are excluded first — secondary records carry no sequence or qualities at all. An alignment whose start lies before the tile was counted in an earlier tile and is skipped, so each read is attributed to exactly one tile.

filter	kept	dropped	share
primary alignments in the sampled tiles	1,011,354		
mapping quality >= 0		0	0.00%
no SA tag (not a split read)		2,127	0.21%
clipped fraction <= 5%		87	0.01%
divergence de <= 0.01		14,165	1.40%
counted	994,975		98.38%

At most 25 reads are taken from any one tile, so the budget spreads over many tiles instead of exhausting the first few.

Mapping quality is not a uniqueness filter here. The target is a *diploid* assembly: a read from a homozygous region maps equally well to both haplotypes and scores MAPQ 0 however perfect its alignment. Requiring MAPQ 60 keeps under 10% of primaries — exactly the haplotype-divergent ones. Measured both ways the gap estimates agree within about 3 standard errors, so the threshold is 0 and the identity, clipping and split-read filters do the work.

Each remaining filter removes alignments whose *path* cannot be believed: a split read is least reliable exactly at its ends, a heavily clipped read is mostly somewhere else, and a read at 3% divergence is evidence about the assembly rather than about sequencing error.

3. What counts as an event

The model's states are M, I and D, so an alignment is already a state path: `100M1I100M` is 100 M columns, one I, then 100 more. `M`, `=` and `X` all map to M — the state does not care whether the base matched, only that read and template both advanced. A run of length n contributes n-1 internal transitions plus one transition into it from the run before.

- **Clips, reference skips and padding break the path** rather than transitioning through it. A clip means the read left the template, which the model's free start and free end already handle.
- **Runs longer than 20 bp split the path.** A 138 bp deletion is a variant, an assembly error or a collapsed repeat, not a sequencing indel, and counting it drives gamma_D towards 1. The long run and the two transitions touching it are dropped and the alignment continues as two independent paths. Dropping the whole read instead would bias the other way, since reads crossing hard sequence carry most of the short gaps too.
- **I<->D adjacencies have no edge in this model.** They are counted and reported, never folded into a neighbour.

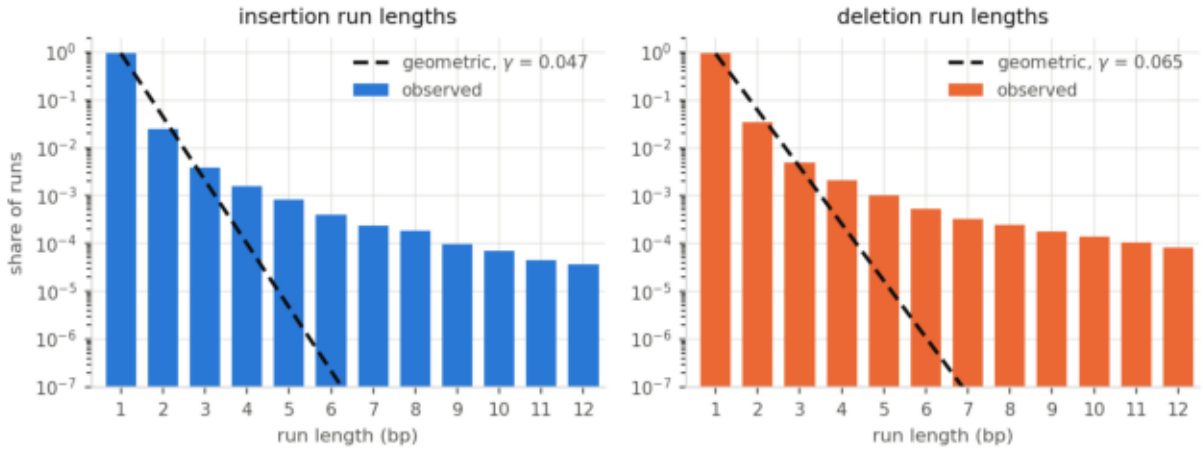
4. What was found

994,975 alignments, 15.3 Gb of aligned columns.

event	count	per aligned base
aligned columns (M)	15,299,116,725	
matches	15,296,947,610	0.999858
mismatches (NM)	2,169,115	1.418e-04
mismatches (base by base)	42,183	1.379e-04
insertion opens (M->I)	6,877,012	4.495e-04
insertion extends (I->I)	338,755	2.214e-05
insertion closes (I->M)	6,877,001	
inserted bases	7,215,767	4.716e-04
deletion opens (M->D)	12,804,095	8.369e-04
deletion extends (D->D)	889,565	5.814e-05
deletion closes (D->M)	12,804,100	
deleted bases	13,693,666	8.951e-04
gap events of either kind	19,681,107	1.286e-03
paths split at a run > 20 bp	5,738	
I<->D adjacencies (no model edge)	6	

Two mismatch figures are given because they are measured differently. The NM figure subtracts every indel base from the aligner's edit distance, over all 994,975 reads. The base-by-base figure compares read to reference directly over a 20,000-read subsample, and is the one to trust: it cannot be confused by an edit distance that also counts something else. The two differ by 3%.

Deletions open 1.9x more often than insertions, and both are short: 95% of insertions and 94% of deletions are a single base.



Observed run lengths against the geometric law the model assumes.

The bars leave the line at the tail. The model can only produce geometric runs, so the excess of long gaps is a misspecification, not an estimation error — it is what context-dependent gap parameters would absorb.

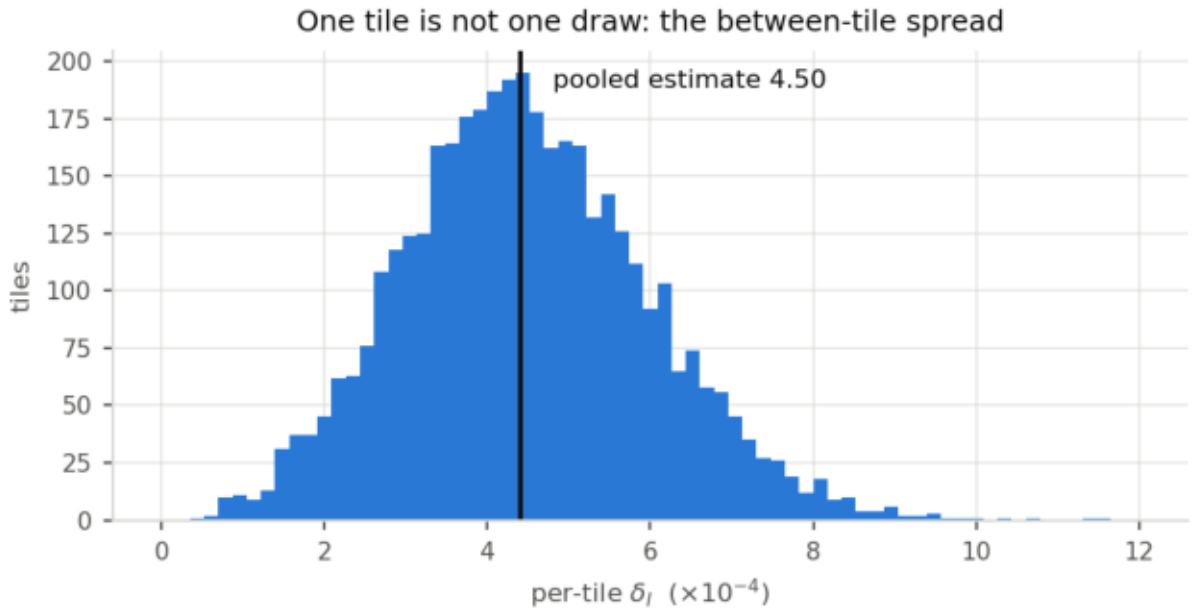
5. The estimates

With the path known, the maximum-likelihood estimate of a transition probability is the ratio of counts — no optimiser, no EM:

```
delta_I = n(M->I) / [n(M->M) + n(M->I) + n(M->D)]      gamma_I = n(I->I) / [n(I->I) + n(I->M)]
delta_D = n(M->D) / [n(M->M) + n(M->I) + n(M->D)]      gamma_D = n(D->D) / [n(D->D) + n(D->M)]
```

parameter	estimate	jackknife	binomial	meaning
delta_I	4.4953e-04	7.8e-07	1.7e-07	insertion opens per M column
delta_D	8.3697e-04	1.3e-06	2.3e-07	deletion opens per M column
gamma_I	0.0469	0.0003	0.0001	mean run 1.05 bp
gamma_D	0.0650	0.0003	0.0001	mean run 1.07 bp

The jackknife is the honest error bar. The binomial standard error treats every aligned base as an independent draw, which they are not: gap rates vary with context and reads in one tile share it. Deleting one tile at a time and re-estimating gives a spread 4.5x wider, over 39,799 tiles.



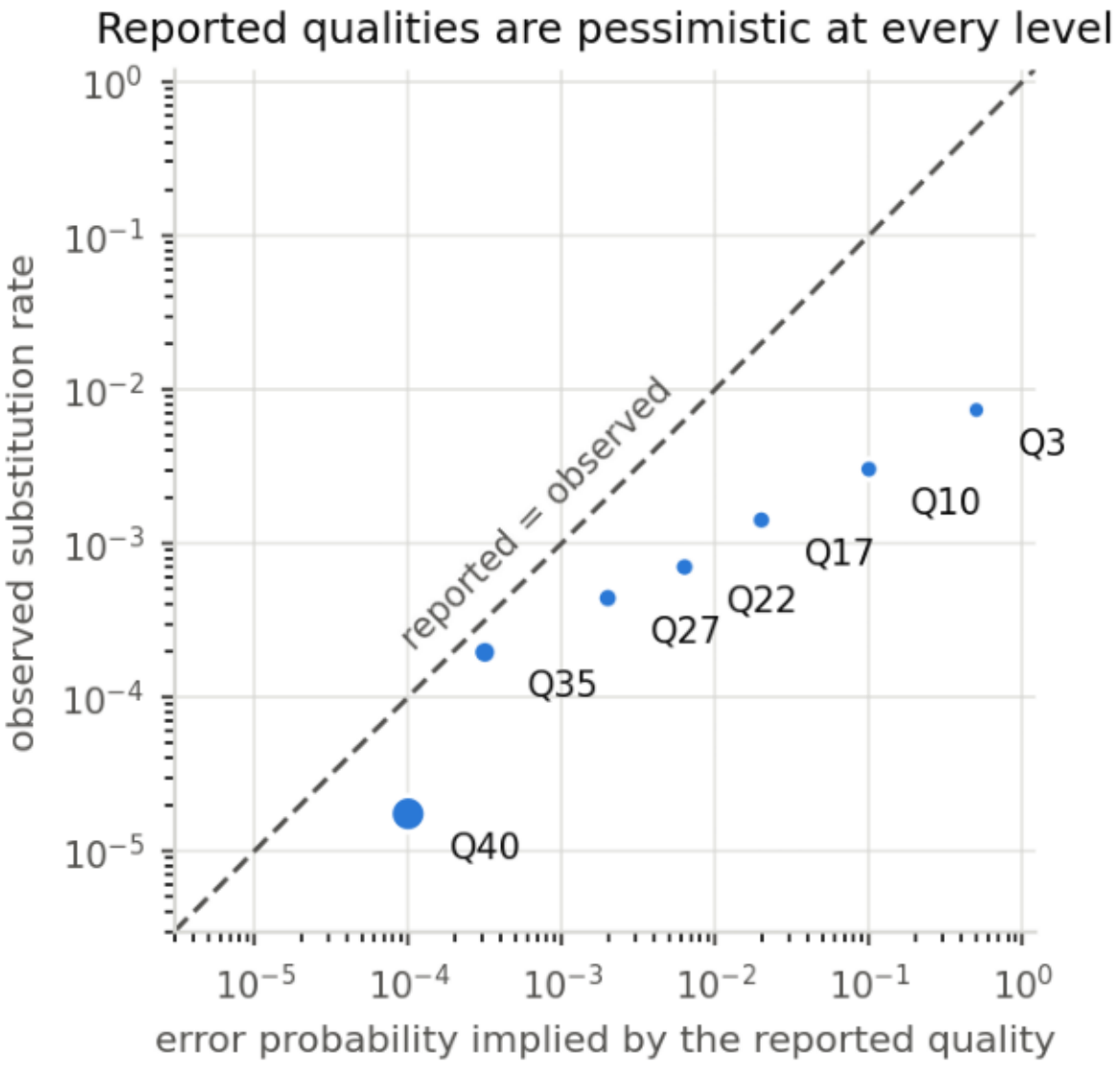
Each tile gives its own estimate; the histogram's width is what the jackknife measures.

Everything else in the model is fixed or derived rather than fitted. M->M is 1 - delta_I - delta_D - tau, B->I is delta_I, I->M is 1 - gamma_I - tau. The termination probability tau is not estimated: read length is conditioned on, and fitting tau to observed lengths would import the length model that conditioning exists to remove. The per-base error probability eps comes from the quality string, not from this fit — which section 6 shows to be a problem.

6. Emission calibration, measured but not applied

The same pass compares read to reference base by base and tallies by reported quality. Reported qualities are pessimistic at every level:

reported	bases	implied eps	observed	as Phred
Q3	1,229,658	5.01e-01	7.34e-03	Q21.3
Q10	3,395,608	1.00e-01	3.03e-03	Q25.2
Q17	4,721,871	2.00e-02	1.41e-03	Q28.5
Q22	5,623,639	6.31e-03	7.00e-04	Q31.5
Q27	7,600,158	2.00e-03	4.39e-04	Q33.6
Q35	22,428,019	3.16e-04	1.95e-04	Q37.1
Q40	260,901,996	1.00e-04	1.74e-05	Q47.6

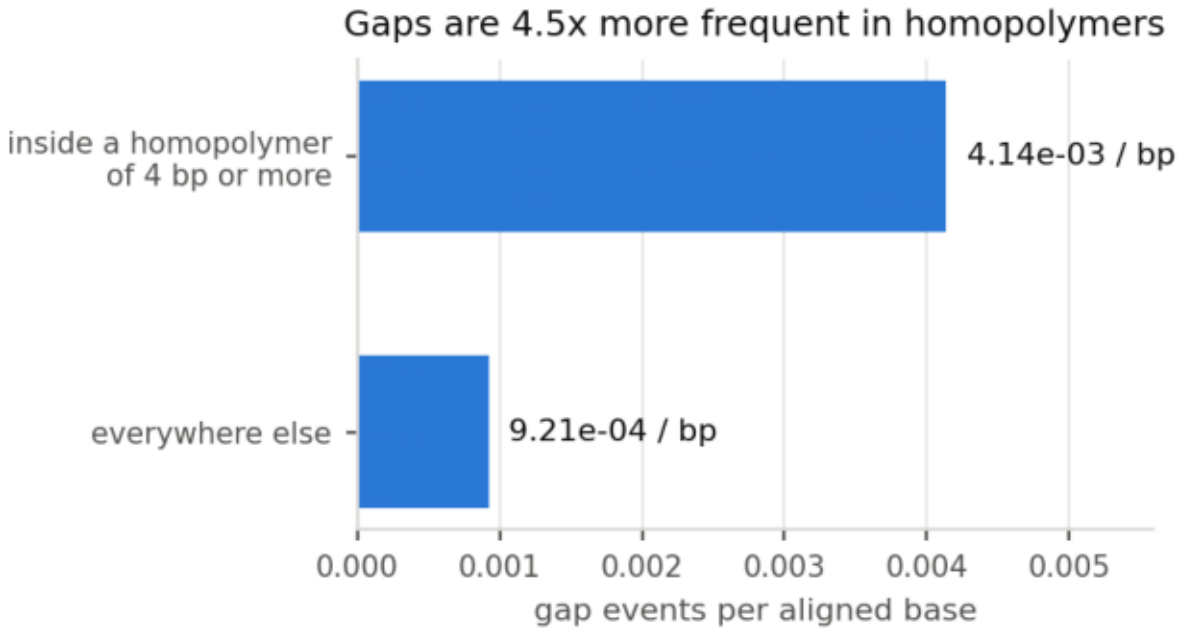


Every point sits below the diagonal: bases are better than they claim.

`eps_from_phred` takes the reported value at face value, so a substitution currently costs log(3(1-eps)/eps) = 10.3 nats where the data say 11.9 — a ~15% under-charge wherever a locus is discriminated by substitutions rather than indels, and a much larger one on low-quality bases, which the model presently treats as nearly uninformative.

This is **not** applied, because the measurement is confounded in both directions: the curated assembly was built from these same reads, so a read error that reached the consensus reads as a match, deflating the observed rate; residual assembly errors and haplotype mismapping read as read errors, inflating it. A clean estimate wants a held-out assembly or variant-masked regions.

7. Context dependence



Gap rate inside homopolymers against everywhere else.

Gaps are 4.5x more frequent inside homopolymers of 4 bp or more (4.14e-03 per base) than elsewhere (9.21e-04 per base). One global gap rate is therefore wrong in a measurable way, and wrong exactly where HiFi indel error concentrates. This is the size of what context-dependent gap parameters would buy, and the reason the seed locus's microsatellite verdict is the least trustworthy magnitude the model produces.

8. What this cannot tell you

- **The path is minimap2's, not the model's posterior.** Its affine penalties (`map-hifi`: O 4,24 / E 2,1) decide which gaps exist at all, and its leftmost convention decides where a tied gap is placed. The estimator itself is checked by a round trip on sampled reads, where the path really is the model's, and recovers the generating parameters within 4 standard errors on all four.
- **The reference is not independent of the reads.** See section 6.
- **20 bp as the gap ceiling is a judgement.** Sequencing indels and small variants overlap in size, and nothing in the data marks the boundary.
- **Fitting against the other assembly moves delta_I by 5%** — 60x the jackknife error. Which assembly is used as the reference, not the sample size, is what limits this estimate.
- **Contigs shorter than one tile are unrepresented.**